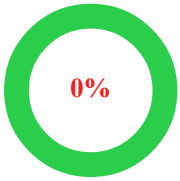
SmallSEOTools

# Plagiarism Detection Report by SmallSEOTOOLS




| | | | |
|---|---|---|---|
| Plagiarism | 0% | Partial Match | 0% |
| Exact Match | 0% | Unique | 100% |

## Scan details

| Total Words | Total Characters | Plagiarized Sentences | Unique Sentences |
|---|---|---|---|
| 383 | 2655 | 0 | 22 (100%) |

### #1 100% Unique

To develop a To-Do List application using React and TypeScript that allows users to add tasks, mark tasks as completed, delete tasks, and view task statistics.

Develop a responsive To-Do List web application using React and TypeScript. The application should allow users to enter new tasks, display the list of tasks, mark tasks as completed using checkboxes, and delete tasks when they are no longer required. The application should also display the total number of tasks, completed tasks, and pending tasks, along with a progress bar indicating the completion percentage.

React is a JavaScript library used to build interactive user interfaces through reusable components. TypeScript is a superset of JavaScript that provides static typing and helps identify errors during development.

In this practical, React and TypeScript are used to create a To-Do List application. The Task interface defines the structure of each task with an ID, title, and completion status. The useState Hook manages the list of tasks and generates unique task IDs. Components such as TaskInput and TaskItem are created to handle task entry and display.

Step for Implementation:

Create a React application using Vite with the TypeScript template.

Install the required dependencies using npm.

Create a types.ts file and define the Task interface.

Create the TaskItem.tsx component to display, complete, and delete tasks.

Develop the App.tsx component to manage the task list using the useState Hook.

Create App.css and index.css to style the application.

Configure main.tsx to render the React application.

Run the application using the Vite development server.

Test adding, completing, and deleting tasks in the browser.

The application successfully performs the required task-management operations. New tasks are added to the list, and checkboxes allow users to mark tasks as completed or pending. The delete button removes unwanted tasks. The total, completed, and pending counts update dynamically whenever the task list changes. The progress bar reflects the percentage of completed tasks. TypeScript interfaces help maintain consistent task data, while React Hooks provide efficient state management and rendering.

https://github.com/juhiparwa-1507/Web-Technology-Lab.git

Thus, a To-Do List application was successfully developed using React and TypeScript. The application demonstrates reusable components, TypeScript interfaces, state management with the useState Hook, event handling, and dynamic rendering. Users can add, complete, and delete tasks while monitoring task statistics and completion progress.